Benchmark: **MA.7.DP.2.2** · Interpret and compare probabilities Student code: _____ Calculator: yes

Learning goal: I can find and interpret the probability of a simple event. (circle your level after scoring)

4	I can compare probabilities of events from different experiments and explain which is more likely.
3	I can write the probability of a simple event as a fraction, decimal, or percent and say what it means.
2	I can tell whether an event is likely, unlikely, certain, or impossible.
1	I need help to start this benchmark.

1. The probability of rain tomorrow is reported as 75%.

Is this event likely or unlikely? Explain using the probability.

A. Likely, because 75% is greater than 50% and closer to 1 than to 0.

B. Unlikely, because 75% is less than 100%, so the event might not happen.

C. Certain, because 75% is so close to 100% that it must happen.

D. As likely as not, because 75% is a percentage and not a fraction.

2. Which value **cannot** be the probability of an event?

A. 0

B. 110%

C. $\frac{7}{8}$

D. 1

3. The probability that Mia makes her next free throw is $\frac{5}{8}$. The probability that Jon makes his next free throw is $\frac{1}{4}$.

Which event is more likely?

A. There is not enough information.

B. Mia makes her next free throw

C. Jon makes his next free throw

D. They are equally likely.

Turn over →

Benchmark: **MA.7.DP.2.2** · Interpret and compare probabilities Student code: _____

4. A probability is given for each event.

Describe each event as likely, unlikely or neither.

Check one box in each row.

	Likely	Unlikely	Neither likely nor unlikely
The probability that it snows in Orlando, Florida, this winter is 2%.	☐	☐	☐
The probability that a spinner with two equal sections, one red and one blue, lands on red is $\frac{1}{2}$.	☐	☐	☐
The probability that a hurricane passes within 100 miles of the city in the next two days is 0.6.	☐	☐	☐

5. A fair 10-sided die has faces numbered 1 through 10. The probability that it rains tomorrow is 0.2.

Which is more likely: that the die shows a number greater than 3, or that it rains tomorrow? Write both probabilities in the same form to justify your answer, and find the difference between the two probabilities.

Score: ____ / 5 My level: 1 2 3 4 → copy both to your progress strip

Answer key — C-DP.2.2 · MA.7.DP.2.2 · Interpret and compare probabilities

Unit 2: Probability. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **A. Likely, because 75% is greater than 50% and closer to 1 than to 0.**

$75\% = 0.75$, which is greater than 0.5, so it is closer to 1 than to 0.

An event with a probability less than 0.5 is unlikely and an event with a probability greater than 0.5 is likely, so this event is likely.

2. Regular **B. 110%**

Every probability is between 0 and 1, inclusive (or between 0% and 100%).

110% cannot be a probability because a probability written as a percent can never be greater than 100%.

3. Regular **B. Mia makes her next free throw**

$\frac{5}{8} \approx 0.63$ and $\frac{1}{4} = 0.25$.

The greater probability is $\frac{5}{8}$, so the event "Mia makes her next free throw" is more likely.

4. Regular **The probability that it snows in Orlando, Florida, this winter is 2%. → Unlikely; The probability that a spinner with two equal sections, one red and one blue, lands on red is $\frac{1}{2}$. → Neither likely nor unlikely; The probability that a hurricane passes within 100 miles of the city in the next two days is 0.6. → Likely**

Write each probability as a decimal and compare it with 0.5: an event with a probability greater than 0.5 is likely, less than 0.5 is unlikely, and exactly 0.5 is neither.

$2\% = 0.02$: less than 0.5, so unlikely.

$\frac{1}{2} = 0.5$: exactly 0.5, so neither.

0.6: greater than 0.5, so likely.

Scoring: 1 point only if every match is correct; anything else earns 0.

5. Rigor **The die shows a number greater than 3; $\frac{7}{10} = 0.7$ and 0.2, so $0.7 > 0.2$, by 0.5 ($\frac{1}{2}$ or 50% also accepted). Score: 1 point — the more likely event with both probabilities in one form and the difference; the event alone earns 0.**

$P(\text{the die shows a number greater than } 3) = \frac{7}{10} = 0.7$ (7 favorable of 10 equally likely outcomes).

$P(\text{it rains tomorrow}) = 0.2$.

$0.7 - 0.2 = 0.5$, so a number greater than 3 is more likely, by 0.5.

Re-check strip (3 items, C-DP.2.2-R)

1. Regular **A. 140%**

Every probability is between 0 and 1, inclusive (or between 0% and 100%).

140% cannot be a probability because a probability written as a percent can never be greater than 100%.

2. Regular **B. The seed sprouts this week**

Write both probabilities in the same form. As decimals: 0.875 and $\frac{7}{10} = 0.7$.

$0.875 > 0.7$, so "the seed sprouts this week" is more likely. A probability can be compared whatever form it is written in.

3. Regular **The probability that a hurricane passes within 100 miles of the city in the next two days is 0.65. → Likely; The probability that a spinner with two equal sections, one red and one blue, lands on red is 50%. → Neither likely nor unlikely; The probability that a thunderstorm is within 5 miles of your house tomorrow is $\frac{3}{10}$. → Unlikely**

Write each probability as a decimal and compare it with 0.5: an event with a probability greater than 0.5 is likely, less than 0.5 is unlikely, and exactly 0.5 is neither.

0.65: greater than 0.5, so likely.

$50\% = 0.5$: exactly 0.5, so neither.

$\frac{3}{10} = 0.3$: less than 0.5, so unlikely.

Scoring: 1 point only if every match is correct; anything else earns 0.

Comparison strip (2 items, C-DP.2.2-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **D. $\frac{6}{5} = 1\frac{1}{5}$**

Every probability is between 0 and 1, inclusive (or between 0% and 100%).

$\frac{6}{5}$ cannot be a probability because a fraction whose numerator is greater than its denominator is greater than 1.

2. Regular **B. Likely, because 85% is greater than 50% and closer to 1 than to 0.**

$85\% = 0.85$, which is greater than 0.5, so it is closer to 1 than to 0.

An event with a probability less than 0.5 is unlikely and an event with a probability greater than 0.5 is likely, so this event is likely.

Reteach plan — C-DP.2.2

MA.7.DP.2.2 · Interpret and compare probabilities

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-DP.2.2-R

Benchmark: **MA.7.DP.2.2** · Interpret and compare probabilities Student code: _____ Calculator: yes

1. Which value **cannot** be the probability of an event?

A. 140%

B. 1

C. $\frac{5}{8}$

D. 10%

2. The probability that the seed sprouts this week is 0.875. The probability that the Tigers win their next game is $\frac{7}{10}$.

Which event is more likely?

A. The forms cannot be compared.

B. The seed sprouts this week

C. The Tigers win their next game

D. They are equally likely.

3. A probability is given for each event.

Describe each event as likely, unlikely or neither.

Check one box in each row.

	Likely	Unlikely	Neither likely nor unlikely
The probability that a hurricane passes within 100 miles of the city in the next two days is 0.65.	☐	☐	☐
The probability that a spinner with two equal sections, one red and one blue, lands on red is 50%.	☐	☐	☐
The probability that a thunderstorm is within 5 miles of your house tomorrow is $\frac{3}{10}$.	☐	☐	☐

Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-DP.2.2-X

Benchmark: **MA.7.DP.2.2** · Interpret and compare probabilities Student code: _____ Calculator: yes

1. Which value **cannot** be the probability of an event?

A. 0

B. $\frac{3}{8}$

C. 0.8

D. $\frac{6}{5}$

2. The probability of a bus arriving on time is reported as 85%.

Is this event likely or unlikely? Explain using the probability.

A. Unlikely, because 85% is less than 100%, so the event might not happen.

B. Likely, because 85% is greater than 50% and closer to 1 than to 0.

C. As likely as not, because 85% is a percentage and not a fraction.

D. Certain, because 85% is so close to 100% that it must happen.

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-DP.2.2-X

Benchmark: **MA.7.DP.2.2** · Interpret and compare probabilities Student code: _____ Calculator: yes

1. Which value **cannot** be the probability of an event?

A. 0

B. $\frac{3}{8}$

C. 0.8

D. $\frac{6}{5}$

2. The probability of a bus arriving on time is reported as 85%.

Is this event likely or unlikely? Explain using the probability.

A. Unlikely, because 85% is less than 100%, so the event might not happen.

B. Likely, because 85% is greater than 50% and closer to 1 than to 0.

C. As likely as not, because 85% is a percentage and not a fraction.

D. Certain, because 85% is so close to 100% that it must happen.

Score: ____ / 2 → **copy to your progress strip**